Example 3.28 Find x such that $-\pi \leq x \leq \pi$ and $\cos 2x = \sin x$

Solution:

We have $\cos 2x = \sin x$ which gives
$2\sin^2 x + \sin x - 1 = 0$. The roots of the equation are
$\sin x = \dfrac{-1 \pm 3}{4} = -1$ or $\dfrac{1}{2}$
Now, $\sin x = \dfrac{1}{2} \Rightarrow x = \dfrac{\pi}{6}, \dfrac{5\pi}{6}$
Also $\sin x = -1 \Rightarrow x = -\dfrac{\pi}{2}$
Thus, $x = -\dfrac{\pi}{2}, \dfrac{\pi}{6}, \dfrac{5\pi}{6}$.

Example 3.29 Find the values of (i) $\sin 18^\circ$ (ii) $\cos 18^\circ$ (iii) $\sin 72^\circ$ (iv) $\cos 36^\circ$ (v) $\sin 54^\circ$

Solution:

(i) Let $\theta = 18^\circ$. Then $5\theta = 90^\circ$
$3\theta + 2\theta = 90^\circ \Rightarrow 2\theta = 90^\circ - 3\theta$
$\sin 2\theta = \sin(90^\circ - 3\theta) = \cos 3\theta$
$2\sin\theta\cos\theta = 4\cos^3\theta - 3\cos\theta$. Since $\cos\theta = \cos 18^\circ \neq 0$, we have
$2\sin\theta = 4\cos^2\theta - 3 = 4(1 - \sin^2\theta) - 3$
$4\sin^2\theta + 2\sin\theta - 1 = 0$
$\sin\theta = \dfrac{-2 \pm \sqrt{4 - 4(4)(-1)}}{2(4)} = \dfrac{-1 \pm \sqrt{5}}{4}$
Thus, $\sin 18^\circ = \dfrac{\sqrt{5} - 1}{4}$ (positive sign is taken. Why ?)

(ii) $\cos 18^\circ = \sqrt{1 - \sin^2 18^\circ}$
$= \sqrt{1 - \left[\dfrac{\sqrt{5} - 1}{4}\right]^2} = \dfrac{1}{4}\sqrt{16 - (5 + 1 - 2\sqrt{5})} = \dfrac{1}{4}\sqrt{10 + 2\sqrt{5}}$

(iii) $\sin 72^\circ = \sin(90^\circ - 18^\circ) = \cos 18^\circ = \dfrac{1}{4}\sqrt{10 + 2\sqrt{5}}$

(iv) $\cos 36^\circ = 1 - 2\sin^2 18^\circ = 1 - 2\left[\dfrac{\sqrt{5} - 1}{4}\right]^2 = \dfrac{\sqrt{5} + 1}{4}$

(v) $\sin 54^\circ = \sin(90^\circ - 36^\circ) = \cos 36^\circ = \dfrac{\sqrt{5} + 1}{4}$.

Observe that $\sin 18^\circ = \cos 72^\circ$, $\cos 18^\circ = \sin 72^\circ$ and $\cos 36^\circ = \sin 54^\circ$

Example 3.30 If $\tan\dfrac{\theta}{2} = \sqrt{\dfrac{1-a}{1+a}}\ \tan\dfrac{\phi}{2}$, then prove that $\cos\phi = \dfrac{\cos\theta - a}{1 - a\cos\theta}$.